# THE SUPPORT SUBCATEGORY EQUALS THE SMALLEST SERRE SUBCATEGORY CONTAINING F CLOSED UNDER EXTENSIONS AND DIRECT SUMS

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

*Source reference.* Original lemma 0GPF: sources/stacks-chow/tags.txt, source lines 20825–20825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $X$ be a quasi-compact and quasi-separated scheme. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. The following two subcategories of $QCoh(\mathcal{O}_X)$ are equal*

1. *the full subcategory $\mathcal{A} \subset QCoh(\mathcal{O}_X)$ whose objects are the quasi-coherent modules whose support is (set theoretically) contained in $Supp(\mathcal{F})$,*
2. *the smallest Serre subcategory $\mathcal{B} \subset QCoh(\mathcal{O}_X)$ containing $\mathcal{F}$ closed under extensions and arbitrary direct sums.*

*Proof.* Observe that the statement makes sense as finitely presented $\mathcal{O}_X$-modules are quasi-coherent. Since $\mathcal{A}$ is a Serre subcategory closed under extensions and direct sums and since $\mathcal{F}$ is an object of $\mathcal{A}$ we see that $\mathcal{B} \subset \mathcal{A}$. Thus it remains to show that $\mathcal{A}$ is contained in $\mathcal{B}$.

Let $\mathcal{E}$ be an object of $\mathcal{A}$. There exists a maximal submodule $\mathcal{E}' \subset \mathcal{E}$ which is in $\mathcal{B}$. Namely, suppose $\mathcal{E}_i \subset \mathcal{E}$, $i \in I$ is the set of subobjects which are objects of $\mathcal{B}$. Then $\bigoplus \mathcal{E}_i$ is in $\mathcal{B}$ and so is

$$\mathcal{E}' = \mathrm{Im}(\bigoplus \mathcal{E}_i \longrightarrow \mathcal{E})$$

This is clearly the maximal submodule we were looking for.

Now suppose that we have a nonzero map $\mathcal{G} \to \mathcal{E}/\mathcal{E}'$ with $\mathcal{G}$ in $\mathcal{B}$. Then $\mathcal{G}' = \mathcal{E} \times_{\mathcal{E}/\mathcal{E}'} \mathcal{G}$ is in $\mathcal{B}$ as an extension of $\mathcal{E}'$ and $\mathcal{G}$. Then the image $\mathcal{G}' \to \mathcal{E}$ would be strictly bigger than $\mathcal{E}'$, contradicting the maximality of $\mathcal{E}'$. Thus it suffices to show the claim in the following paragraph.

Let $\mathcal{E}$ be an nonzero object of $\mathcal{A}$. We claim that there is a nonzero map $\mathcal{G} \to \mathcal{E}$ with $\mathcal{G}$ in $\mathcal{B}$. We will prove this by induction on the minimal number $n$ of affine opens

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$U_i$ of $X$ such that $\text{Supp}(\mathcal{E}) \subset U_1 \cup \ldots \cup U_n$. Set $U = U_n$ and denote $j : U \to X$ the inclusion morphism. Denote $\mathcal{E}' = \text{Im}(\mathcal{E} \to j_*\mathcal{E}|_U)$. Then the kernel $\mathcal{E}''$ of the surjection $\mathcal{E} \to \mathcal{E}'$ has support contained in $U_1 \cup \ldots \cup U_{n-1}$. Thus if $\mathcal{E}''$ is nonzero, then we win. In other words, we may assume that $\mathcal{E} \subset j_*\mathcal{E}|_U$. In particular, we see that $\mathcal{E}|_U$ is nonzero. By Lemma 0GPF there exists a nonzero map $\mathcal{F}|_U \to \mathcal{E}|_U$. This corresponds to a map

$$\varphi : \mathcal{F} \longrightarrow j_*(\mathcal{E}|_U)$$

whose restriction to $U$ is nonzero. Setting $\mathcal{G} = \varphi^{-1}(\mathcal{E})$ we conclude. $\square$